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```
# LEVEL 4 - TASK 4.1
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("Route_Journey_Duration.csv")

df["Journey_Duration"] = pd.to_numeric(
    df["Journey_Duration"],
    errors="coerce"
)

route_duration = df.groupby("Route_Number")["Journey_Duration"].mean().reset_index()

route_duration = route_duration.sort_values("Journey_Duration")

plt.figure(figsize=(12,6))

plt.bar(
    route_duration["Route_Number"].astype(str),
    route_duration["Journey_Duration"]
)

plt.title("Average Journey Duration Across Different Routes")
plt.xlabel("Route Number")
plt.ylabel("Average Journey Duration (Minutes)")
plt.xticks(rotation=90)

plt.grid(axis="y", linestyle="--", alpha=0.5)

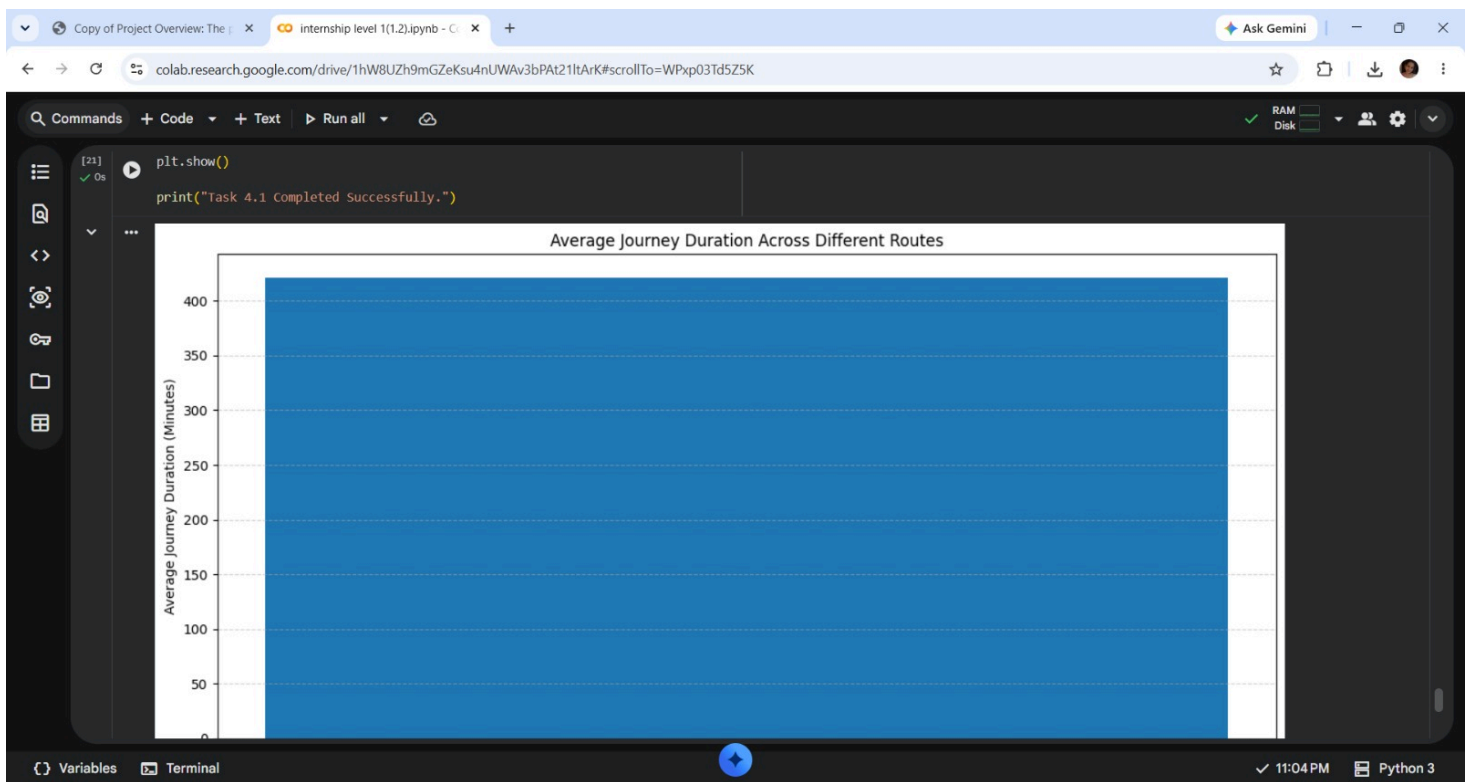
plt.tight_layout()

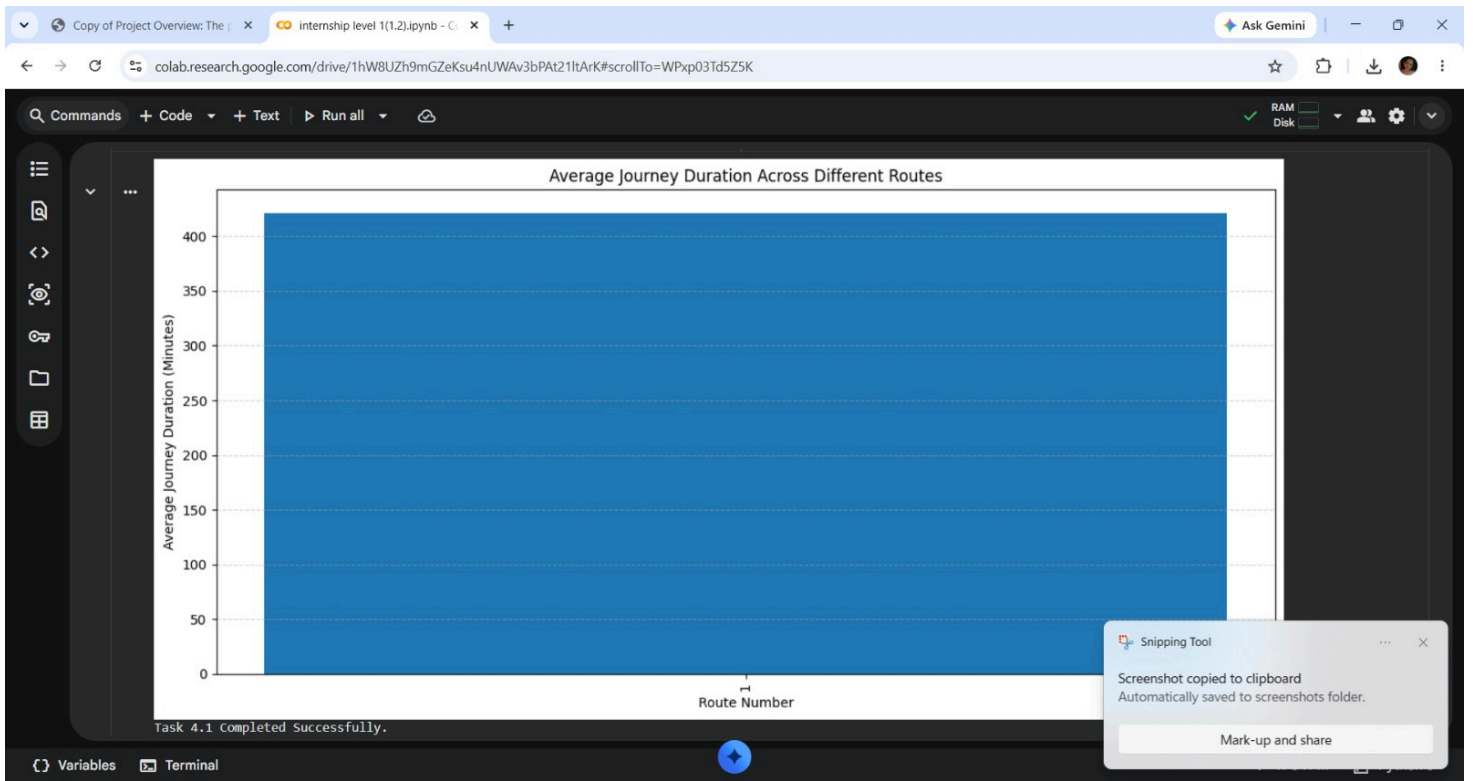
plt.savefig("Task4_1_Journey_Duration_Across_Routes.png")
```

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```
# LEVEL 4 - TASK 4.2
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("Station_Traffic.csv")

df = df.sort_values("Train_Traffic", ascending=False)

top10 = df.head(10)

plt.figure(figsize=(12,6))

plt.bar(top10["Station_Name"], top10["Train_Traffic"])

plt.title("Top 10 Stations with Highest Train Traffic")
plt.xlabel("Station Name")
plt.ylabel("Number of Trains")

plt.xticks(rotation=45, ha="right")

plt.grid(axis="y", linestyle="--", alpha=0.5)

plt.tight_layout()

plt.savefig("Task4_2_Top10_Station_Traffic.png")

plt.show()

bottom10 = df.tail(10)

plt.figure(figsize=(12,6))
```

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plt.xticks(rotation=45, ha="right")

plt.grid(axis="y", linestyle="--", alpha=0.5)

plt.tight_layout()

plt.savefig("Task4_2_Top10_Station_Traffic.png")

plt.show()

bottom10 = df.tail(10)

plt.figure(figsize=(12,6))

plt.bar(bottom10["Station_Name"], bottom10["Train_Traffic"])

plt.title("Top 10 Stations with Lowest Train Traffic")
plt.xlabel("Station Name")
plt.ylabel("Number of Trains")

plt.xticks(rotation=45, ha="right")

plt.grid(axis="y", linestyle="--", alpha=0.5)

plt.tight_layout()

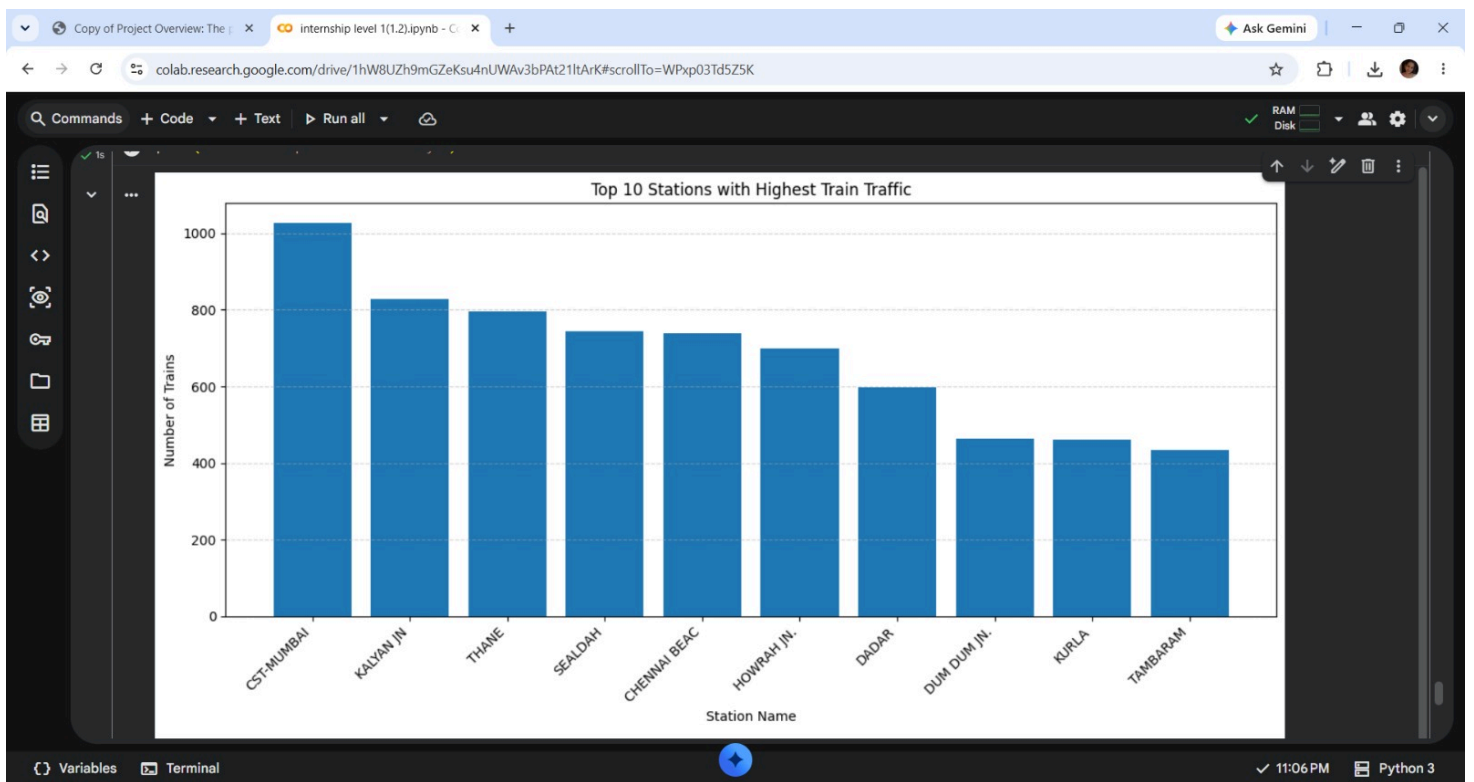
plt.savefig("Task4_2_Bottom10_Station_Traffic.png")

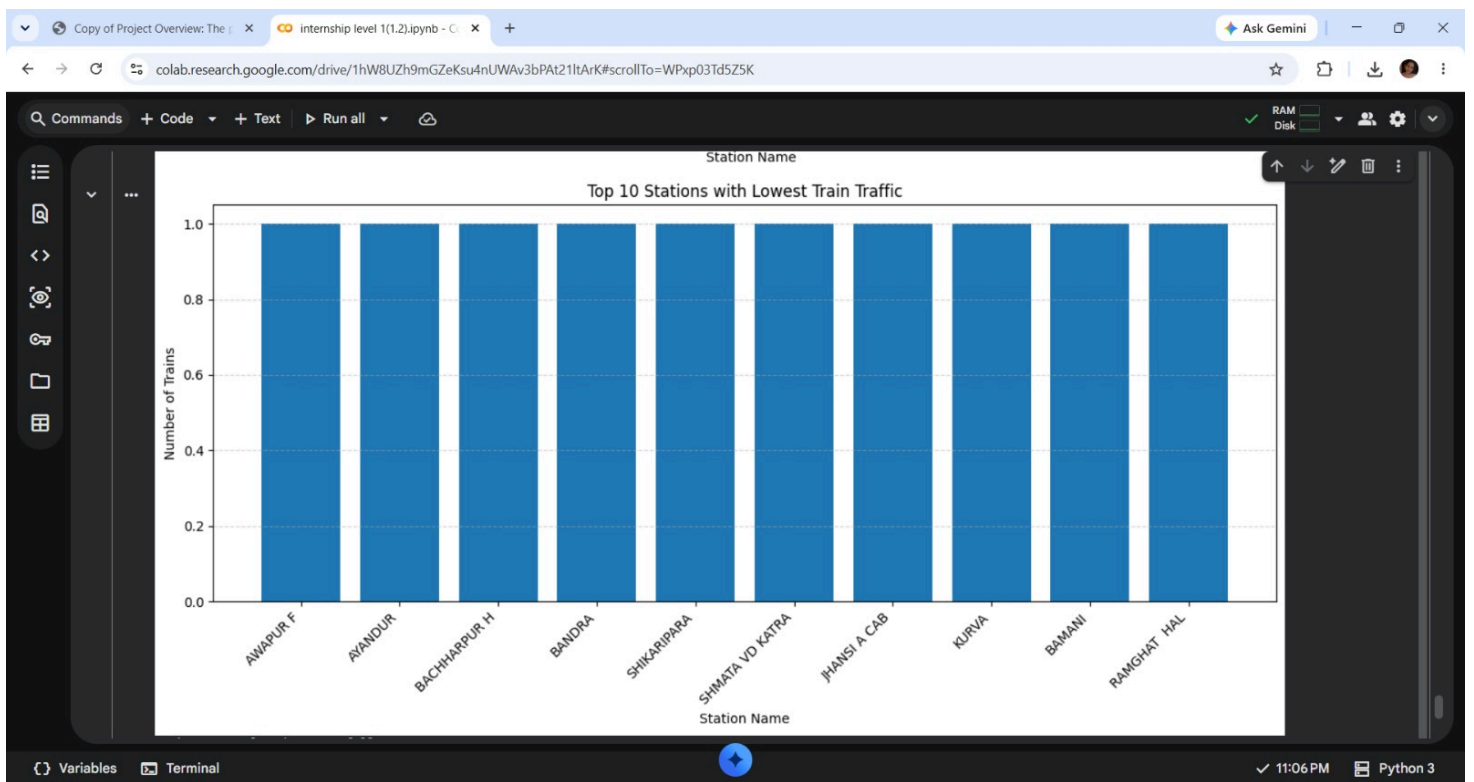
plt.show()

print("Task 4.2 Completed Successfully.")
```

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```
# LEVEL 4 - TASK 4.3
import pandas as pd
import matplotlib.pyplot as plt

features = pd.read_csv("Feature_Engineered_Dataset.csv")
journey = pd.read_csv("Journey_Duration.csv")

journey["Journey_Duration"] = pd.to_timedelta(
    journey["Journey_Duration"],
    errors="coerce"
).dt.total_seconds() / 60

data = pd.merge(features, journey, on="Train_No")

plt.figure(figsize=(10,6))

plt.scatter(
    data["Total Distance"],
    data["Journey_Duration"]
)

plt.title("Distance vs Journey Duration")
plt.xlabel("Total Distance (km)")
plt.ylabel("Journey Duration (Minutes)")

plt.grid(True)

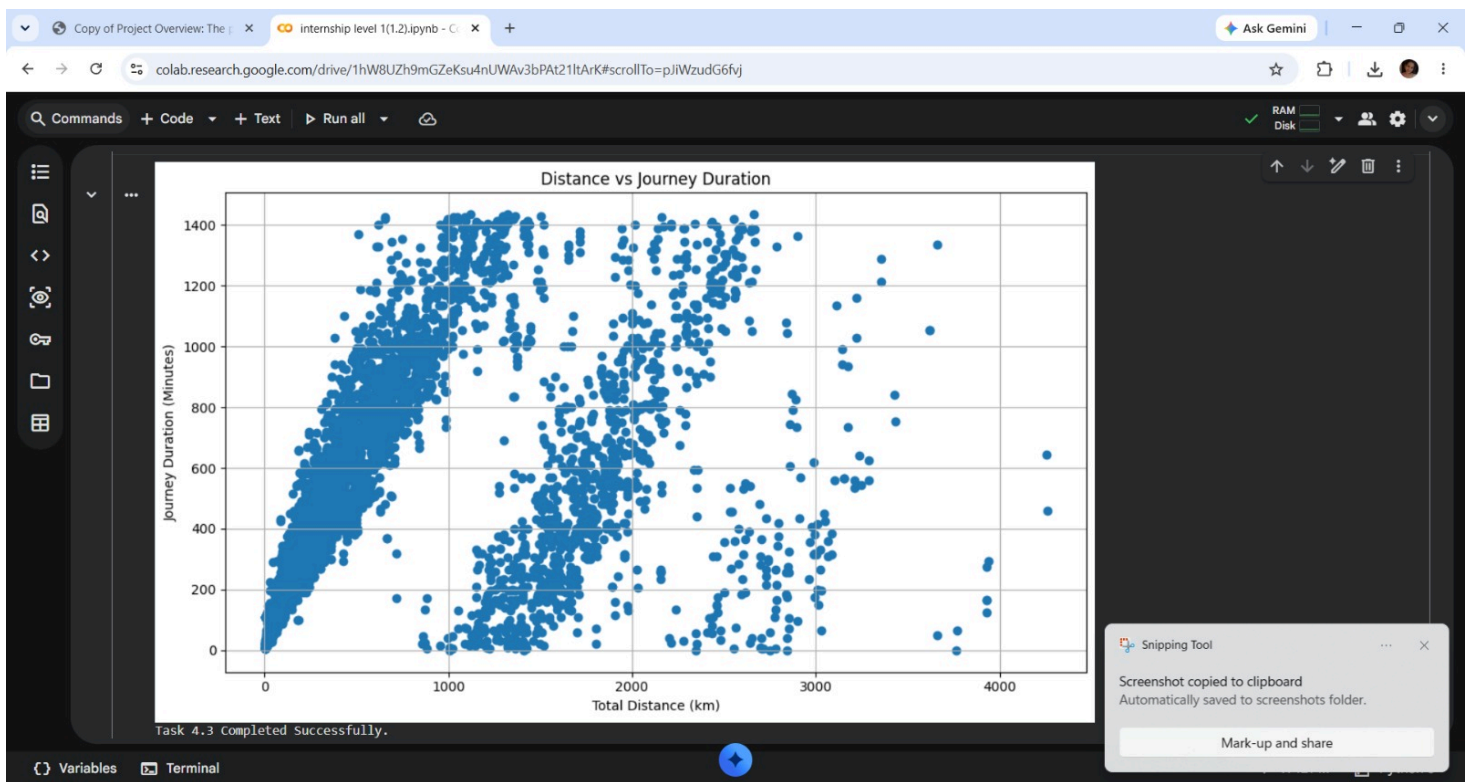
plt.tight_layout()

plt.savefig("Task4_3_Distance_vs_Journey_Duration.png")

plt.show()
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```
# LEVEL 4 - TASK 4.4
print("===== PATTERN ANALYSIS =====\n")

print("1. Journey duration varies across different train routes.")
print("2. Longer routes generally require more travel time.")
print("3. Some stations have significantly higher train traffic than others.")
print("4. Major junctions experience higher train movement compared to smaller stations.")
print("5. There is a positive relationship between total distance and journey duration.")
print("6. As the travel distance increases, the journey duration also tends to increase.")
print("7. The visualizations help identify important trends and support further analysis.")

print("\nTask 4.4 Completed Successfully.")
```

```
... ===== PATTERN ANALYSIS =====

1. Journey duration varies across different train routes.
2. Longer routes generally require more travel time.
3. Some stations have significantly higher train traffic than others.
4. Major junctions experience higher train movement compared to smaller stations.
5. There is a positive relationship between total distance and journey duration.
6. As the travel distance increases, the journey duration also tends to increase.
7. The visualizations help identify important trends and support further analysis.

Task 4.4 Completed Successfully.
```

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